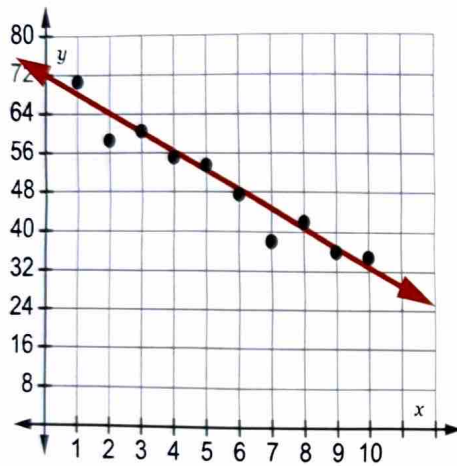


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Practice Test

Amani is calculating the slope of the trend line in the scatterplot shown.



How can Amani use the trend line to check the reasonableness of her answer?

She should expect the slope to be positive because all the plotted points have positive coordinates.

She should expect the slope to be negative because more points are below the line than above the line.

She should expect the slope to be positive because the line has a positive y -intercept.

She should expect the slope to be negative because the line tilts down moving from left to right on the graph. ✓

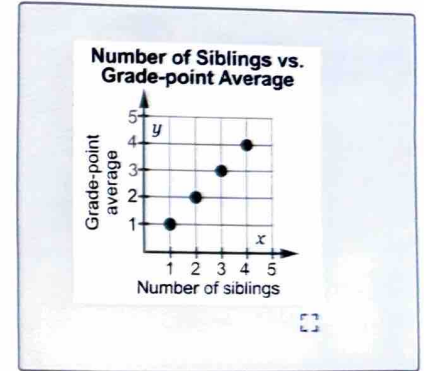
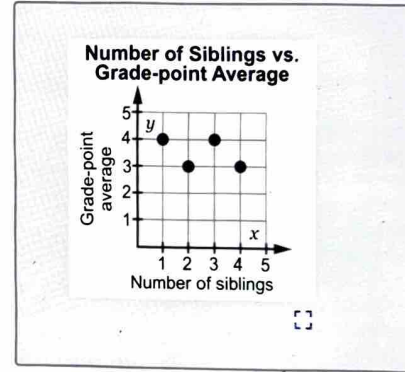
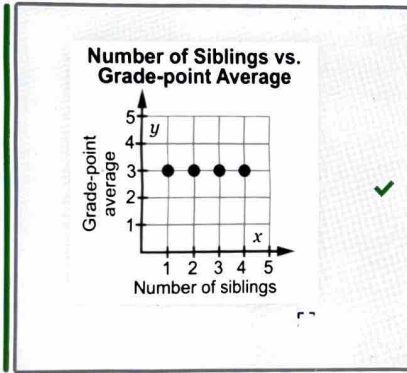
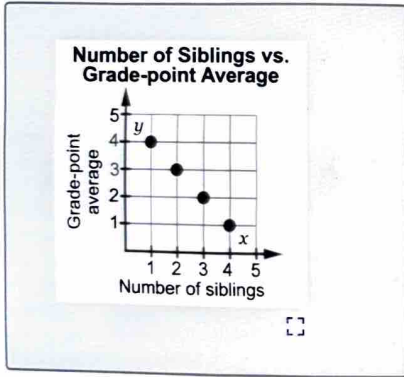
RETURN TO
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SUMMARY

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79%
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Practice Test

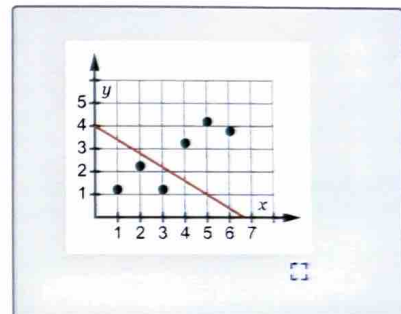
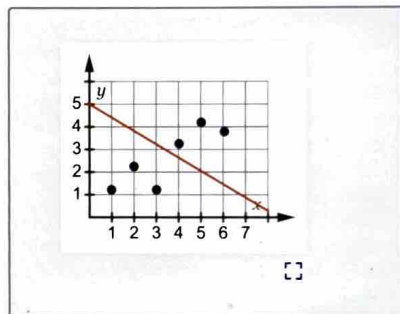
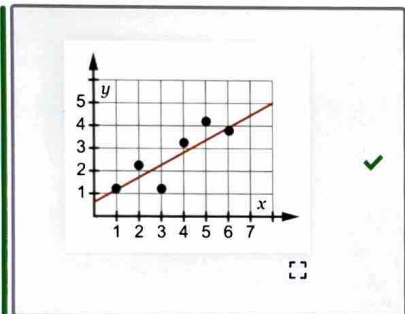
Wyatt wants to study the relationship between the number of siblings that a student has and the student's grade-point average using a scatterplot. He believes that if the points form a straight line, then there is a relationship between the two variables. Which scatterplot proves that Wyatt's statement is not always true?



RETURN TO
ASSESSMENT
SUMMARY

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SCORE

[RETURN TO ASSESSMENT SUMMARY](#)



79%
SCORE



Pruma made the given two-way table showing her friend's favorite fruit juices.

Favorite Juices			
	Grapefruit Juice	Orange Juice	Total
Girls	7		16
Boys	3		14
Total	10		

Which shows the correct values for Pruma's friends who prefer orange juice?

- girls: 9; boys: 11 ✓
- girls: 23; boys: 17
- girls: 11; boys: 9 *
- girls: 17; boys: 23

RETURN TO ASSESSMENT SUMMARY

79% SCORE



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Practice Test



Esther made the given two-way table showing the number of her schoolmates who have different pets.

Students' Pets			
	Cats	Dogs	Total
7th Grade			
8th Grade	26	35	61
Total	54	48	

Which shows the correct values for Esther's seventh-grade classmates who have pets?

- cats: 13; dogs: 28
- cats: 28; dogs: 13 ✓
- cats: 83; dogs: 80
- cats: 80; dogs: 83

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ASSESSMENT
SUMMARY

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79%
SCORE



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Practice Test



Which statement is true about the purpose of trend lines in scatterplots?

A trend line must pass through the largest data point.

A trend line must pass through the smallest data point.

A trend line can be used to make predictions in real-world situations. ✓

A trend line shows the exact data points collected.

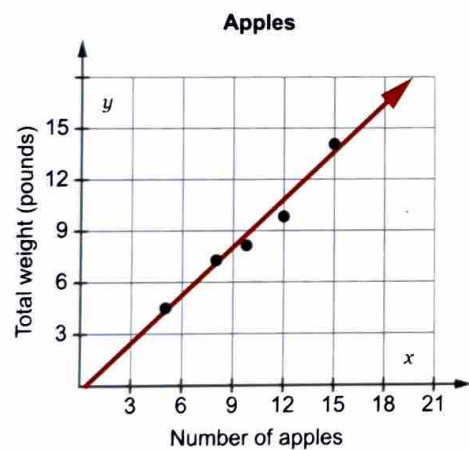
RETURN TO ASSESSMENT SUMMARY

★ 79%
SCORE

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Practice Test

Naomi recorded the total weight of different numbers of apples. Her results are shown by using the scatterplot and the trend line shown.



For which number of apples is the prediction of the total weight an extrapolation?

18



6

12

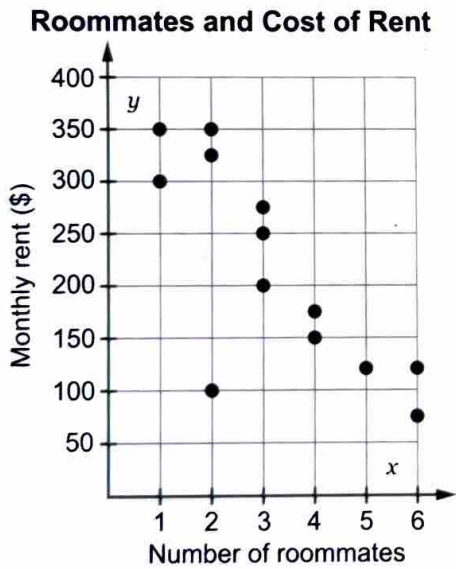
15

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ASSESSMENT
SUMMARY

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The scatterplot shows the number of roommates and the monthly cost of rent for a group of college students.



Which statement about the scatterplot is true?

- Including the point (2, 100) could cause the description of the data set to be understated. ✓
- Including the point (2, 100) could cause the description of the data set to be overstated.
- Although (2, 100) is an extreme value, it should be part of the description of the relationship between the number of roommates and monthly rent.
- The point (2, 100) shows that there is no relationship between the number of roommates and monthly rent.

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★ 79% SCORE



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Practice Test



The table shows a set of data.

Data	
x	y
1.6	39
1.9	38
2.3	42
3.4	40
3.8	41
4.2	44
4.3	42
4.6	45
4.8	44

Which statement about the table is true?

There is a cluster, and as x increases, y increases. ✓

There is a cluster, and as x decreases, y increases.

There is not a cluster, and as x increases, y increases.

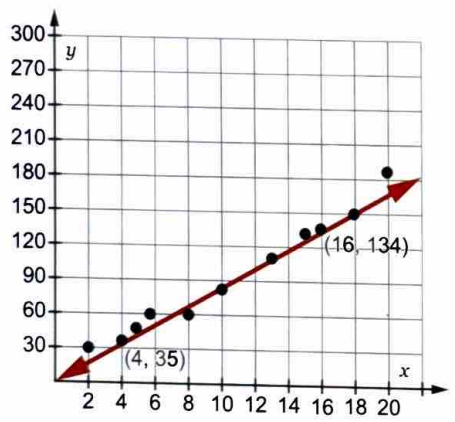
There is not a cluster, and as x decreases, y increases.

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Practice Test



Which expression can be simplified to find the slope of the line of best-fit in the scatterplot shown?

☒ $\frac{134-35}{16-4}$

☐ $\frac{4-16}{35-134}$

☐ $\frac{4-16}{134-35}$

☐ $\frac{134-16}{35-4}$

RETURN TO ASSESSMENT SUMMARY

★ 79%
SCORE

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- X
- X

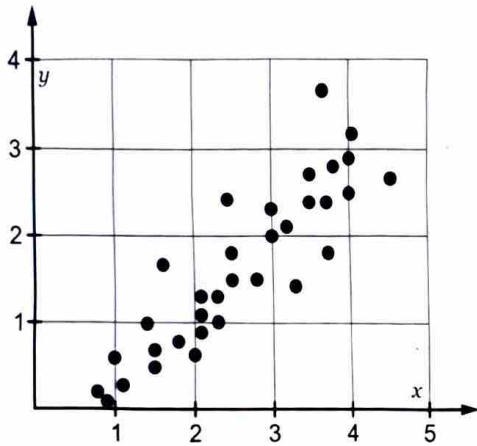
Practice Test



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Which characteristics describe the scatterplot? Choose three correct answers.

☒ strong association ✗

☐ weak association ✓

☒ linear ✓

☒ positive correlation ✓

☐ nonlinear

☐ negative correlation

RETURN TO
ASSESSMENT
SUMMARY

★ 79%
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✗

Practice Test



1-14 of 14

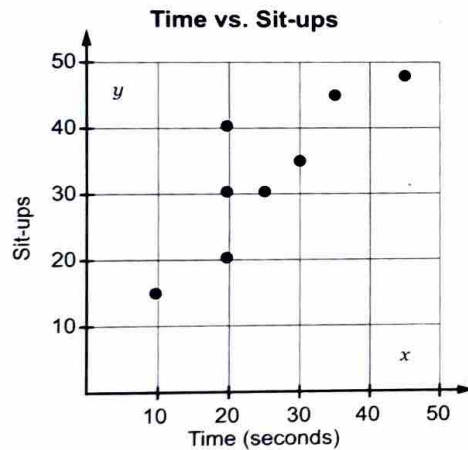
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The given scatterplot was created from a table of values and represents the relationship between the number of seconds spent doing sit-ups and the number of sit-ups completed for some of the members of a physical education class.



Assuming that none of the data points overlap and that all the data points were plotted, how many data points were in the table?

 ✓

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ASSESSMENT
SUMMARY

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SCORE



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- 11
- ✗
- ✗

Practice Test

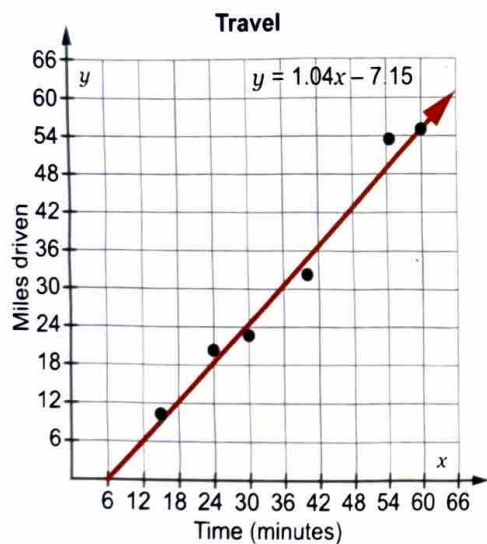


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Practice Test

The points (15, 10), (25, 20), (30, 22), (40, 32), (55, 53), (60, 55) represent the time, x , it takes 6 people to travel y miles. The scatterplot here represents the situation.



Based on the information given, how many miles should a person who drives for 48 minutes expect to travel?

32 miles

42 miles ✓

54 miles ✗

48 miles

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Practice Test



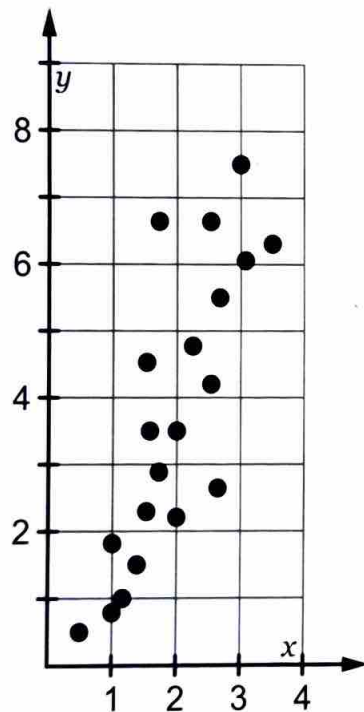
◀ 1-14 of 14 ▶

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What type of association is shown by the scatterplot?

linear, weak



nonlinear, strong

linear, strong



nonlinear, weak

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SUMMARY79%
SCORE

Which would be a point on the scatterplot that represents touchdown passes based on these games played?

 $(2, 4)$

(3, 1)

(1, 2)

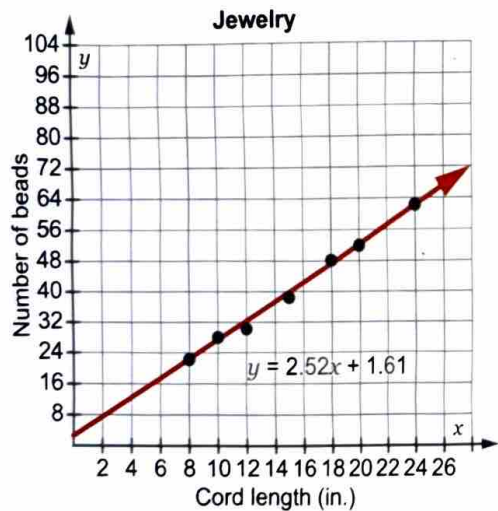
(5, 3) ✓

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ASSESSMENT
SUMMARY

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SCORE

2026 Course 3 - Grade 8 Mathematics
Patterns in Bivariate Data

Angelique makes beaded jewelry. The graph shows the number of beads used for varying cord lengths.



Angelique has a cord that is 32 inches long. Based on the information given, which is the **best** estimate for the number of beads she can use?

12

132

82

68

RETURN TO
ASSESSMENT
SUMMARY

96%
SCORE

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Test



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Patterns in Bivariate Data

Victoria created the scatterplot based on the data in the table for the ages and heights of some teachers in her school.

Teacher Age vs. Height		
Teacher	Age	Height (in.)
1	36	62
2	28	70
3	50	60
4	44	72
5	58	68
6	24	65

She wants to see if a teacher's height depends on his or her age. What did she do wrong when she created the scatterplot?

She mixed up the x - and y -coordinates of the point representing teacher 3. ✓

She plotted the point (36, 62) when she shouldn't have.

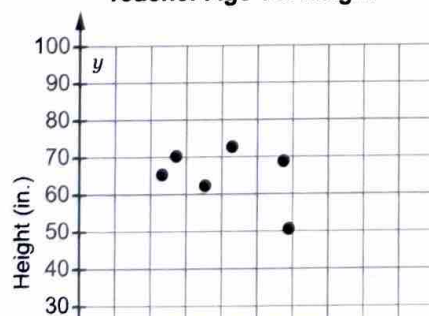
She mixed up the independent and dependent variables.

She labeled the x -axis of the scatterplot "Age" when she should have labeled it "Teacher."

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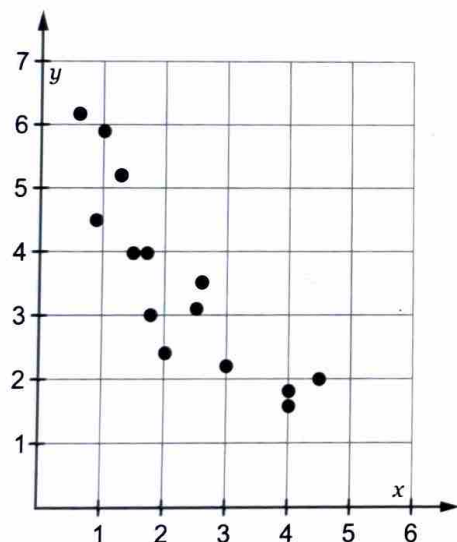
Teacher Age vs. Height





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Patterns in Bivariate Data



Which describes the correlation shown in the scatterplot?

More points are needed to determine the correlation.

There is no correlation in the data set.

There is a negative correlation in the data set. ✓

There is a positive correlation in the data set.

RETURN TO
ASSESSMENT
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Test



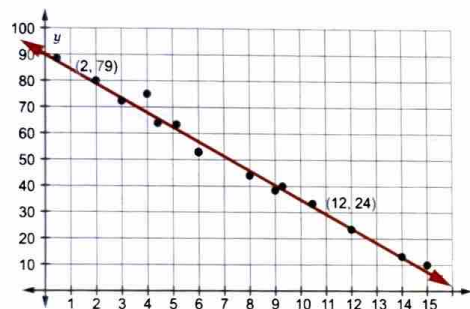
1-25 of 25

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Patterns in Bivariate Data



Which expression can be simplified to find the slope of the trend line in the scatterplot?

$$\frac{24-79}{12+2}$$

$$\frac{24-79}{12-2}$$



$$\frac{24+79}{12+2}$$

$$\frac{24-79}{2-12}$$

RETURN TO
ASSESSMENT
SUMMARY



96%
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- 10
- 11
- X
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- 24
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Test

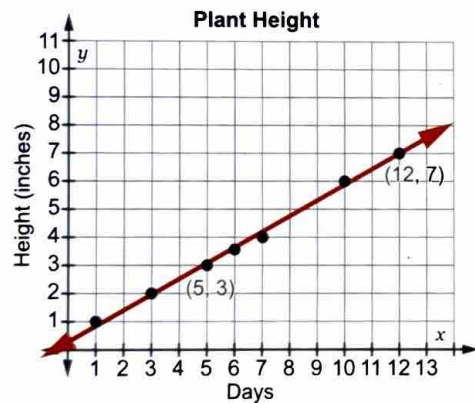


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Patterns in Bivariate Data

A scientist created a scatterplot to display the height of a plant over a 12-day period.



Which is the equation of the trend line that is shown?

$$y = \frac{4}{7}x - \frac{1}{7}$$

$$y = \frac{1}{7}x + \frac{16}{7}$$

$$y = \frac{4}{7}x + \frac{1}{7}$$

$$y = \frac{1}{7}x + \frac{4}{7}$$

RETURN TO
ASSESSMENT
SUMMARY

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Test



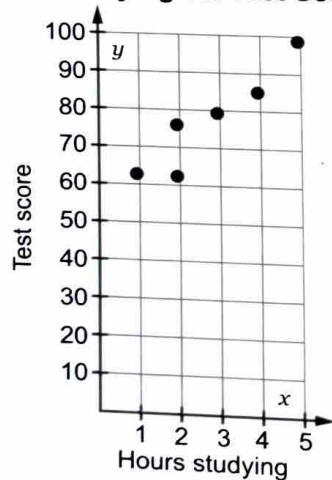
1-25 of 25

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Patterns in Bivariate Data

Mackenzie investigated the relationship between the number of hours studying and the test score for some of the students in her class. She constructed the scatterplot.

Hours Studying vs. Test Score



Which statement is correct based on the scatterplot?

There is a relationship between study hours and test score. As the hours of studying increase, the test scores decrease.

There is no relationship between hours studying and test score because there are multiple test scores for 2 hours studying.

There is a relationship between study hours and test score. As the hours of studying increase, the test scores increase. ✓

There is no relationship between hours studying and test score because the points do not form a straight line.

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Test

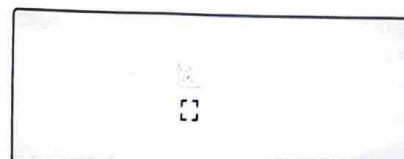
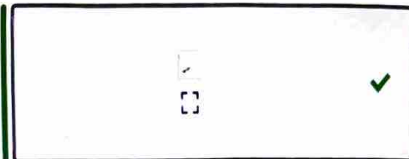
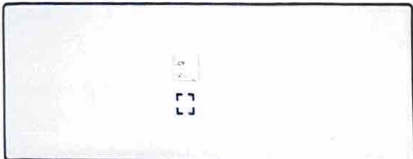
1-25 of 25

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Patterns in Bivariate Data

Which scatterplot displays an example of a cluster?

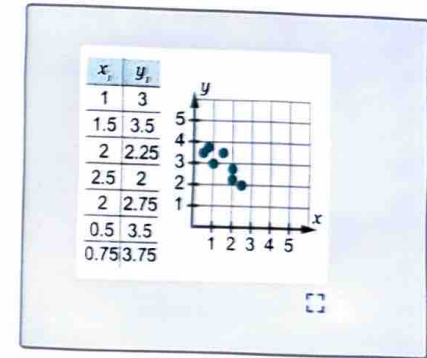
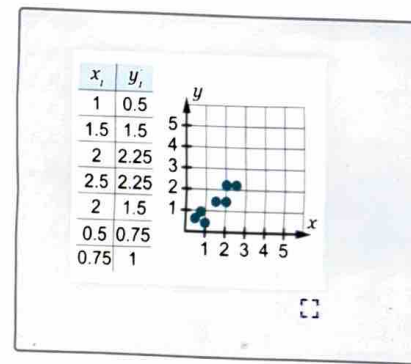
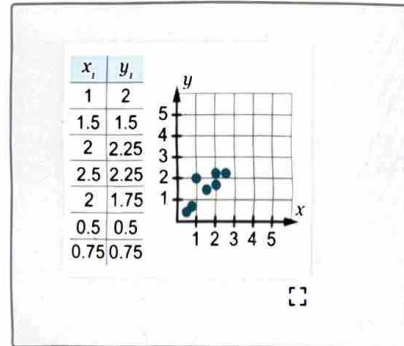
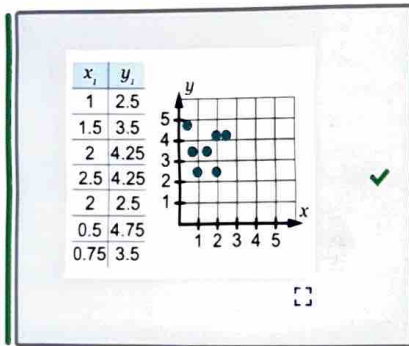


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Patterns in Bivariate Data

Which of the following scatterplots do not show a clear relationship and would not have a trend line?



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Patterns in Bivariate Data

Mrs. Suyata assigned her third- and fourth-period math classes into two study groups. Which describes the variables in the two-way table?

Group Assignments			
	Group 1	Group 2	Total
3rd Period	18	14	32
4th Period	15	17	32
Total	33	31	64

study group number and
period 3

period number and group 1

study group number and
total

period number and study
group number ✓

RETURN TO
ASSESSMENT
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SCORE

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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Patterns in Bivariate Data



The equation $y = 9.56x + 495.35$ represents the number of calories, y , consumed in one day based on a person's weight, x . Jill consumes 1850 calories daily. Jaxon weighs 120 pounds. Which is an accurate comparison?

Jill consumes about 20 more calories than Jaxon.

Jaxon consumes about 20 more calories than Jill.

Jill weighs about 20 pounds more than Jaxon. ✓

Jaxon weighs about 20 pounds more than Jill.

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Test



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Patterns in Bivariate Data

AR

Loren solved the equation $10 = \frac{19}{9}(149) + b$ for b as part of her work to find the equation of a trend line that passes through the points $(1, 130)$ and $(10, 149)$. What error did Loren make?

She should have solved
 $149 = \frac{19}{9}(10) + b$ for b .

She should have
 $1 = \frac{19}{9}(130) + b$ solved
for b .

She should have solved
 $130 = \frac{9}{19}(1) + b$ for b . ❌

She should have solved
 $10 = \frac{9}{19}(149) + b$ for b .

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The data set would have a nonlinear association.

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Patterns in Bivariate Data

Which describes a correct way to construct a scatterplot?

When analyzing the number of students that ride the school bus each day of the week, the day is the dependent variable.

When analyzing the number of home runs that a baseball player hit over several seasons, the season is the independent variable. ✓

When analyzing the number of umbrellas that are sold during days with varying amounts of rain, the number of umbrellas that are sold is the independent variable.

When analyzing the number of registered voters by age, age is the dependent variable.

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96%
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Patterns in Bivariate Data

Yvonne made the given two-way table showing students' favorite fruits.

Favorite Fruits			
	Apples	Bananas	Total
Class A	13	20	33
Class B		16	34
Total	31	37	

According to the table, how many students in class B prefer apples?

16

18 ✓

13

20

RETURN TO
ASSESSMENT
SUMMARY

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Test

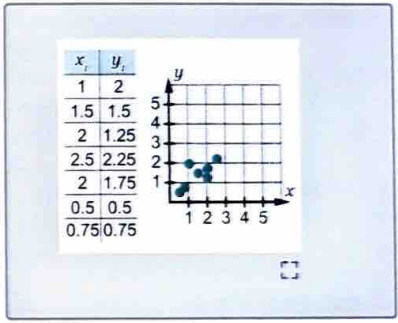
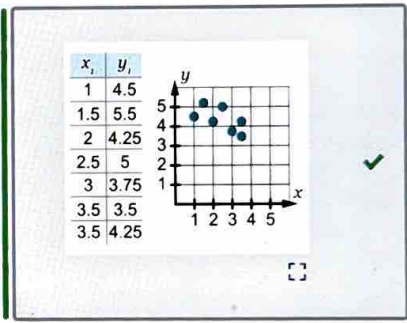
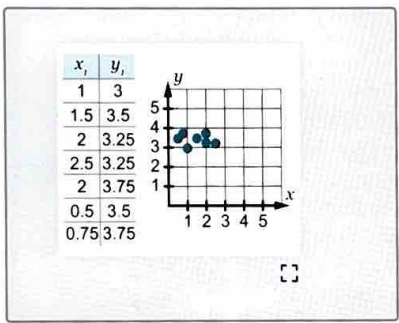
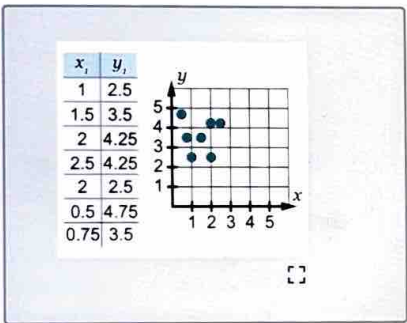


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Which of the following scatterplots would have a trend line with a negative slope?



RETURN TO ASSESSMENT SUMMARY
96% SCORE

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Patterns in Bivariate Data

Which statement is true about a trend line?

A trend line represents only the largest data points.

A trend line represents only the smallest data points.

A regression line and a trend line are opposite terms.

A regression line and trend line are equivalent terms. ✓

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ASSESSMENT
SUMMARY

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SCORE

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Test



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Patterns in Bivariate Data

Subash created the given two-way table.

Teams			
	Team A	Team B	Total
Boys	15	18	33
Girls	17		31
Total	32	32	

According to the table, how many girls were on team B?

14



18

15

17

RETURN TO
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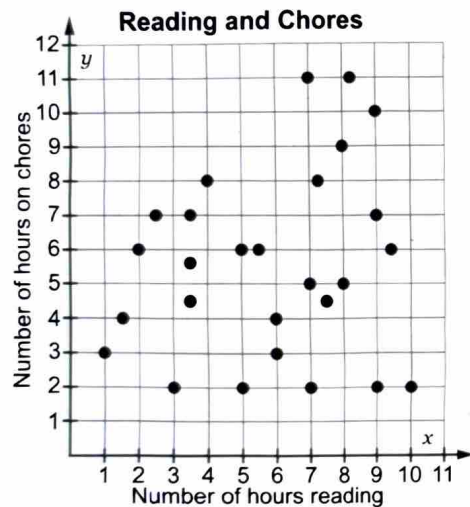
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Patterns in Bivariate Data

The scatterplot shows the number of hours that students read weekly and the number of hours that they spend on chores weekly.



Which statement about the scatterplot is true?

In general, the number of hours of reading does not affect the number of hours used for chores.

As the number of hours of reading increases, the number of hours spent on chores increases.

In general, the number of hours spent on chores is equal to the number of hours spent on reading.

As the number of hours of reading increases, the number of hours spent on chores decreases.

RETURN
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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Test

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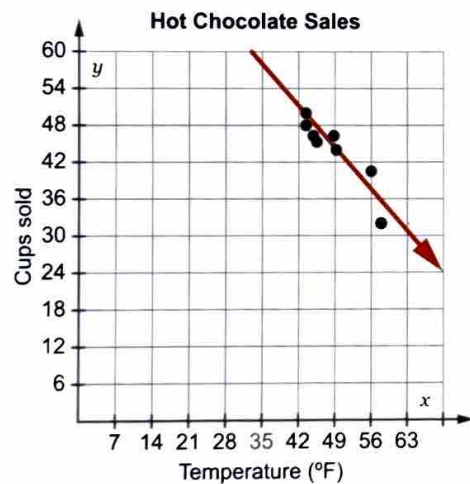


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Patterns in Bivariate Data



Antonia recorded the number of cups of hot chocolate that were sold at soccer games compared to the outside temperature. The graph given represents her data.



For which temperature would the prediction of the number of cups sold be an interpolation?

63° F

35° F

21° F

49° F

RETURN TO
ASSESSMENT
SUMMARY

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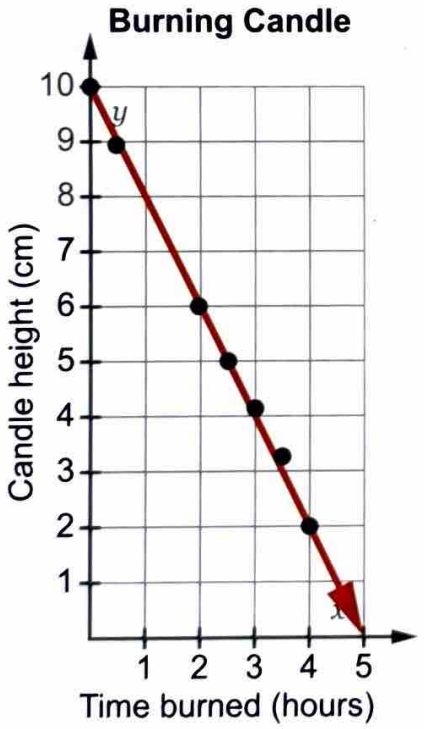


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Submitted



Carina records the height of a candle after x hours burning. Her data are shown in the given graph.



Which is the **best** estimate for the height of the candle after 1 hour?

- 4 cm
- 9 cm
- 5 cm
- 8 cm ✓

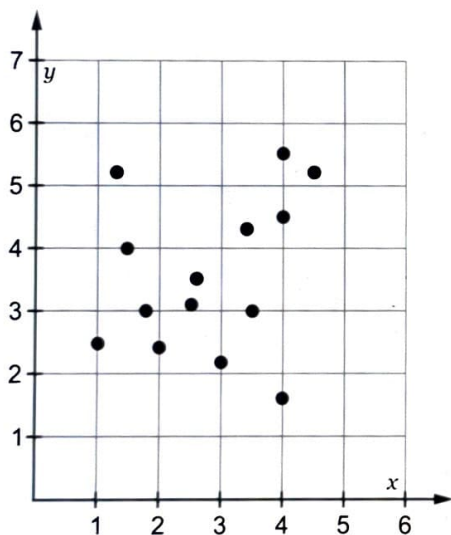
RETURN TO ASSESSMENT SUMMARY

★ 96% SCORE



2026 Course 3 - Grade 8 Mathematics

Patterns in Bivariate Data



Which describes the correlation shown in the scatterplot?

More points are needed to determine the correlation.

There is a negative correlation in the data set.

There is a positive correlation in the data set.

There is no correlation in the data set. ✓

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Test



1-25 of 25

Submitted

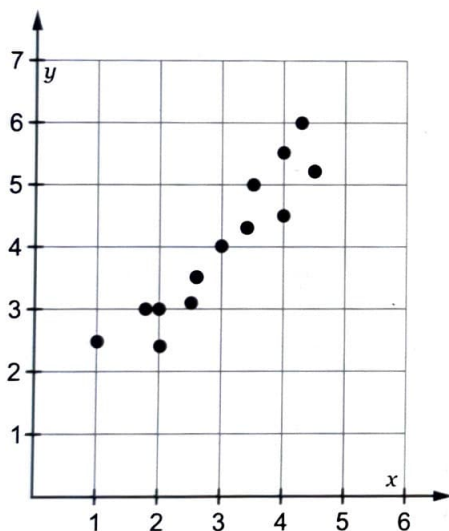


2026 Course 3 - Grade 8 Mathematics

Patterns in Bivariate Data



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Which describes the correlation shown in the scatterplot?

There is a negative correlation in the data set.

There is no correlation in the data set.

More points are needed to determine the correlation.

There is a positive correlation in the data set. ✓

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Test



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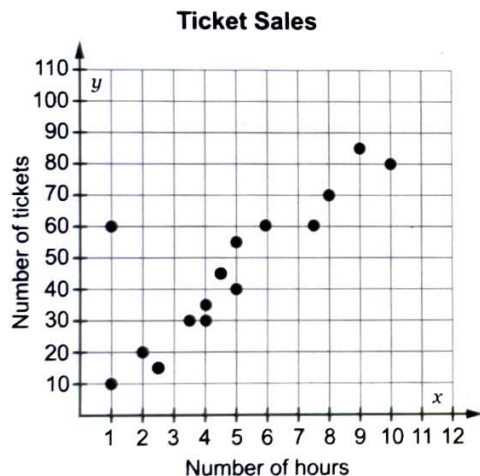
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Patterns in Bivariate Data



The scatterplot shows the number of hours that several students worked to sell tickets for a concert and the number of tickets that each student sold.



Which data value would cause the interpretation of the data set to be overstated?

(10, 80)

(1, 60) ✓

(1, 10)

(9, 85)

RETURN TO
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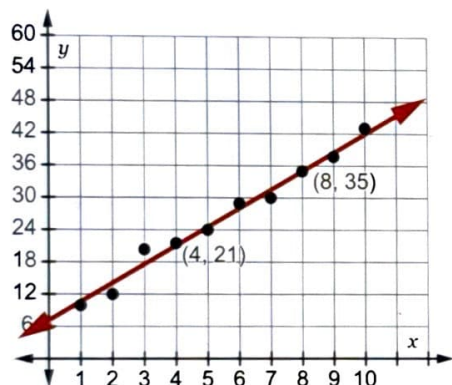


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Patterns in Bivariate Data



Which expression can be simplified to find the slope of the trend line in the scatterplot?

$$\frac{35-21}{8-4}$$



$$\frac{21-35}{8-4}$$

$$\frac{8-4}{21-35}$$

$$\frac{8-4}{35-21}$$

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Test



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